

Sooner or Later?

Promising Delivery Speed in Online Retail

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Online retailers have to provide customers with an estimate of how fast an order can be delivered before they purchase it. This **delivery speed** promise is what online retailers can strategically adjust at almost no cost but what may fundamentally impact business outcomes. It influences consumers' purchasing decisions and post-purchase experiences, often in the opposite direction. On one hand, **an aggressive (i.e., faster) delivery estimate could assure more customers of meeting their deadlines and thus may increase their purchases ex-ante.** On the other hand, an aggressive estimate tends to overpromise customers, risking a longer than expected wait time, which can lower customer satisfaction and increase product returns ex-post. In this research, we study the causal effect of retailers' delivery speed promise on customer behaviors and business performance. Collaborating with Collage.com, an online retailer that sells customized photo products, we co-designed its new shipping policy. We **exogenously** varied the disclosed delivery speed estimates across cities while keeping the physical delivery speed unchanged. Using a difference-in-differences identification and a dataset with 212,340 transactions in 7,090 cities, we find that a one-day faster promise increases sales by 0.73%, profits by 2.0%, and value per order by 3.5%; a one-day slower promise reduces sales by 0.51%, profits by 2.7%, and value per order by 3.1%. However, the aggressive disclosure increases product returns. Retailers could leverage our insights to customize delivery promises for certain products or customers.

Key words: delivery speed information, online retail

1. Introduction

Online retail has continued to thrive and expand at an impressive rate. E-commerce sales in the US accounted for 10.7% of the total retail sales in the first quarter of 2019 and are expected to grow three times faster than traditional retail (MarketingCharts 2019). Online retailers show customers various information about their products and services, such as prices, reviews, inventory availability, and delivery speed. How to effectively leverage such information becomes a strategic decision for online retailers to **manage the customer experience**. For example, the delivery service

is a critical decision factor in online shopping; **shipping speed** has been shown to affect the decisions of 87% of online shoppers, and speed is deemed more important than price when there is a deadline (Chao 2016, MH&L 2016). As consumers increasingly value delivery, not only do speed and reliability become a key battleground in e-commerce but how to disclose information about delivery to customers at the time of purchase also becomes a winning strategy (Mottl 2018).

Retailers take two steps to manage delivery: infrastructure and structure. After deciding on the level of delivery speed and reliability in serving customers, retailers can invest in infrastructures like facilities and transportation to achieve the pre-determined quality. In addition to the existing infrastructures, retailers may leverage structural choices like how to promise delivery speed to further improve the customer experience. Past research has empirically evaluated infrastructural choices, such as shortening the physical delivery time (Fisher et al. 2016) and having high-quality delivery options (Cui et al. 2019a). The literature, however, lacks empirical studies on the structural choices, such as how to manage this information while not changing the physical delivery process. We contribute to the literature by studying the information aspect of delivery speed and its impact.

In practice, the physical delivery speed is always subject to uncertainty due to production, inventory, scheduling, or transportation factors. For example, custom-made products may encounter a longer manufacturing time due to seasonality, and carriers may wait for trucks to be fully loaded in order to ship in large batches, which makes customers' waiting time uncertain. In such circumstances, how to promise customers the delivery speed becomes a strategic lever to manage their expectations. McKinsey has reported that many companies overlook consumer behavior when delivering services and thus recommend that companies pay attention to customer expectations (DeVine et al. 2012, Bhattacharjee et al. 2016). Therefore, it is important to understand consumers' reactions to delivery speed information, as they help retailers decide on their optimal information disclosure policy.

In this paper, we study whether and to what extent consumers respond to online retailers' promised delivery speed. In particular, we study how an aggressive promise and a conservative promise affect customers' purchasing and product return behaviors as well as the company's profitability.

When a retailer chooses an aggressive disclosure, the faster delivery promise can meet more customers' deadlines or demand for speedy delivery, leading to more purchases. The conservative disclosure in delivery speed, however, indicates a longer wait and attracts less customers. Once customers have placed an order, they form an expectation about how long they will have to wait based on the delivery information received. When a retailer promises an aggressive delivery speed and, at the same time, keeps the actual speed unchanged, customers are likely to experience a longer than expected wait time. This might hurt their purchasing experience, cause more returns,

and, in the worst case, discourage their future purchases. The conservative disclosure, on the other hand, might improve a customer’s experience because their order might arrive in advance. In short, retailers need to balance this trade-off between the customer’s expected and experienced delivery speed.

In order to test this, we partnered with Collage.com, a leading online retailer that sells custom photo products in the US. Collage operates as a retail platform by connecting customers with 13 vendors capable of producing customized items. The retail platform enables customers to upload photos, design picture layouts, adjust backgrounds, and edit texts to create custom products and gifts, such as mugs, blankets, pillows, and canvases. The retailer partners with third-party carriers to deliver products from vendors to customers. The company used to have a speed disclosure policy where customers received the same delivery estimates across all cities regardless of how close they were to the vendors. In 2016, in an internal email to the executives, the company’s CEO expressed his concerns that their shipping rule was too inflexible and that he was thinking about making it more customized:

We were actually being inflexible in our initial shipping rules. We will make our estimates much more customized.

— Joe Golden, Co-founder and CEO of Collage.com, 2016

We worked with Collage to design a new shipping disclosure policy that was implemented on December 9th, 2016: **customers in different cities were given different delivery speed estimates based on how close they were to the vendors.** The delivery speed estimates became faster in 13.5% of cities, slower in 58.5% of cities, and did not change in the remaining 27.9% of cities. This creates two groups of cities under the influence of the aggressive disclosure and the conservative disclosure, respectively, as well as a control group without changes in speed estimates. Because all shipping is outsourced to third-party carriers, and the retailer did not alter its physical delivery setup, this new policy only changed the delivery speed information and what consumers expected of the delivery speed. In this way, the policy change works as a quasi-experiment and creates exogenous variations for identifying the causal effect of delivery speed information.

We collected Collage’s transaction-level dataset with 212,340 transactions from 462 SKUs, which spanned from September 2016 to March 2017. We also aggregated the data to create a panel structure over 7,090 cities across 24 weeks. The data includes the standard order information (i.e., the purchasing value of an order, the price of each purchased item, how much customers paid for shipping, and the profit margin of each order), how satisfied customers felt with the purchasing experience and the delivery of their orders (i.e., satisfaction rate and score), and product return information (i.e., whether the product was returned). We study five outcome variables that measure sales and expenditures: **the number of orders, profits, order values, item prices, and shipping**

expenditures. Additionally, we study three outcome variables that measure customers' experiences that could indicate their future retention: product return, satisfied or not, and satisfaction score. In addition, we define the treatment variable as the number of days the retailer promises faster or slower speed under the new disclosure policy, relative to the promise under the old disclosure policy.

We leveraged the quasi-experiment by using the difference-in-differences (DID) identification. We exploited the temporal and geographical variations in the market outcomes of the cities treated with the aggressive disclosure (i.e., cities that received a faster delivery estimate), the cities treated with the conservative disclosure (i.e., cities that received a slower delivery estimate), and untreated cities (i.e., cities that received an unchanged delivery estimate). Our treatment is a continuous change in the delivery promise, ranging from 5.5 days faster to 6 days slower. We verified that the treated and control products followed the same trend before the policy change.

We find that the change in the disclosure strategy significantly affects consumers' purchasing behavior. In particular, when the retailer promises customers one day faster shipping, sales increase by 0.73%, profits increase by 2.0%, and customers spend 3.5% more on each order. These could indicate a significant business impact for the studied company; by varying the promised speed by only one day, Collage would receive 30,000 more orders and 9 million dollars more in revenue.¹ However, when the retailer promises customers one day slower shipping, sales decrease by 0.51%, profits decrease by 2.7%, and customers spend 3.1% less on each order. We also show that the aggressive disclosure increases product returns because consumers are likely to wait longer than expected, whereas the conservative disclosure does not significantly affect returns. Moreover, the conservative disclosure lowers consumer satisfaction, and the aggressive disclosure does not significantly affect consumer satisfaction. These results indicate that in our research context, promising customers a faster delivery speed can boost sales and profitability but at the cost of a higher product return rate; retailers should take caution when trying slower delivery promises because customers may make less purchases, buy cheaper products, and become less happy with the purchasing process.

Our findings indicate that customers are in fact very sensitive to the delivery speed promise when they make purchases online. It is important to note that customers can decide on whether to purchase and to return based on how retailers promise the delivery speed. If online retailers manage customers' expectations properly, by promising the right delivery speed for the right products to the right customers, they will be able to drastically improve their profitability and customer retention without investing heavily in logistics infrastructures.

¹ We computed these order and revenue increases by using Collage's 2019 performance. In 2019, Collage received 4.3 million orders, and its revenue reached 146 million dollars.

2. Literature Review

Our paper is related to the stream of literature on the value of service time, especially in terms of delivery speed and reliability. The past empirical studies have examined the effects of physical delivery speed or quality. For example, a speedy service achieved by a faster drive-thru at a fast-food restaurant (Allon et al. 2011), a shorter queue at a store (Lu et al. 2013), and an increased staffing level (Musalem et al. 2016a) have been shown to improve sales. Consumers' sensitivity in delivery time might vary across online and offline channels (Lim et al. 2020). When the delivery is reliable, products are less likely to be returned (Rao et al. 2014). Customers not only learn about the delivery reliability but also apply their updated beliefs when making future purchasing decisions (Musalem et al. 2016b).

Our work is closely related to Fisher et al. (2016) and Cui et al. (2019a), who showed the benefits of a quick and high-quality delivery to retail platforms. For example, Fisher et al. (2016) demonstrated that when the physical delivery speed becomes faster, total revenue increases by 4%, and net profit increases by 2.2% for an online retailer. Additionally, Cui et al. (2019a) found that when a high-quality delivery option was removed from Alibaba's retail platform, sales dropped by 14%; when it was resumed, sales went back to normal. In their research, the studied companies changed their physical facility network and, as a result, shortened or lengthened their actual delivery time. Our paper, however, focuses on the information side of decisions. The information strategy can be executed at almost zero cost, and at the same time, can profoundly impact consumer choices. Our research complements this literature by examining an important but understudied dimension of delivery service—how to strategically promise the delivery speed in online retail environments.

This paper sheds light on a key market control—information sharing—for retail platforms (Brynjolfsson and Smith 2000). Various forms of information have been shown as effective tools to boost sales and profits in online retail, for example, by sharing timely price information with sellers (Parker et al. 2016), revealing product availability to induce herding and scarcity behaviors (Cui et al. 2019b, Calvo et al. 2019), showing transparent shipping stages for customers' purchases (Bray 2019), and providing both online and offline inventory availability to create an integrated omni-channel shopping experience for customers (Gallino and Moreno 2014). Research has shown that if customers learn about the shipping options after incurring the search cost, they become less sensitive to the shipping price (Dinlersoz and Li 2006). Driven by this fact, several studies have proposed strategies to mitigate consumer price sensitivity by hiding or delaying valuable information to lock customers in to a purchase (Baye and Morgan 2001, Ellison and Ellison 2009, Moon et al. 2018). However, such strategies may lead to negative outcomes because the lack of availability of shipping information in early stages is one of the main reasons why consumers abandon their shopping carts. This indicates that managing shipping information is a critical and nontrivial

decision in online retail. We complement this literature by **purporting** that delivery speed information, when its trade-offs are carefully leveraged, can attract demand, encourage expenditure, and enhance customer retention.

3. Hypothesis Development

Next, we theorize how consumers respond to delivery speed information when shopping online. When deciding whether or not to buy a product, customers often assess the delivery service—how fast the retailer can deliver the product. Shipping speed is deemed as a key factor that affects consumer shopping decisions; sometimes it is more important than price when there is a deadline, such as buying a holiday gift.

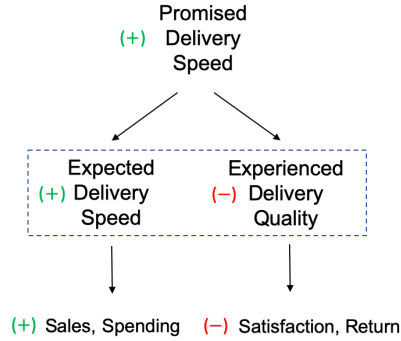
The promised delivery speed and the actual delivery speed jointly determine the customer experience. The shipping speed promised to customers is how fast customers expect to receive the product in the purchasing phase, based on which customers determine whether or not to place an order. As a result, the delivery promise can affect market outcomes, such as sales, profits, expenditures, and the types of products purchased. After an order is placed, it will take some lead time to be delivered, which includes manufacturing, customizing, packaging, and transporting time. In practice, the lead time is always uncertain. Therefore, the actual arrival time is often not exactly what the seller promised to its customers, as it is sometimes faster and sometimes slower. The promised delivery speed, in turn, affects how customers perceive delivery quality in the delivery phase. As a result, the delivery promise can affect market outcomes, such as customer satisfaction and product returns.

Online retailers can strategically choose to aggressively or conservatively promise the delivery speed. An aggressive disclosure means promising a faster shipping speed, and a conservative disclosure means promising a slower shipping speed. We summarize our theoretical framework in Figure 1, which illustrates how delivery information strategies shape consumers' expected and experienced delivery speeds and thereby affect market outcomes. **Figure 1 (a) demonstrates the impact of the aggressive disclosure, and Figure 1 (b) shows the opposite impact for the conservative disclosure.**

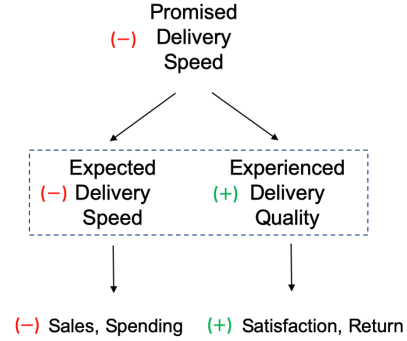
The promised speed helps customers form expectations about how fast they will receive their orders. The more aggressively the retailer promises its delivery speed, the faster customers expect the delivery. Therefore, a fast-delivery promise is likely to meet more demand for a speedy delivery or to meet more customers' deadlines, which will lead to more purchases. In addition, a fast-delivery promise is able to attract more last-minute customers, who tend to be less price-sensitive and willing to buy more expensive products; therefore, the aggressive disclosure can increase customers' spending on each order. Because the shipping fee is proportional to the total spending, customers also spend more on shipping. In contrast, the conservative disclosure strategy promises customers

Figure 1 The Research Framework

(a) Aggressive Delivery Speed Promise



(b) Conservative Delivery Speed Promise



a longer delivery time, which might fail to satisfy some customers, especially those experiencing time pressure or a last-minute need, leading to less purchases and less spending.

HYPOTHESIS 1. *A faster delivery promise will increase the number of orders and customer expenditures per order. A slower delivery promise will reduce the number of orders and customer expenditures per order.*

Customers care about retailers' ability to deliver a product on time. On-time delivery means supplying orders by the promised due date. Customers consider a delayed arrival beyond the expected waiting time as a loss and an early arrival as a positive utility. When a retailer promises an aggressive delivery speed while not changing the actual delivery speed, the seller may miss many delivery due dates. As a result, customers might miss their deadlines or exert extra efforts to track their packages, making them unsatisfied and likely to return the product. Customers might even stop considering the seller as a reliable retailer and, in the worse case, no longer shop on it. A conservative disclosure, however, can improve a customer's experience because the order might arrive in advance. As a result, a company's ability to deliver faster than promised adds value to the purchasing experience, making customers more satisfied and less likely to return the product.

HYPOTHESIS 2. *A faster delivery promise will increase product returns. A slower delivery promise will reduce product returns.*

4. Research Background and Data

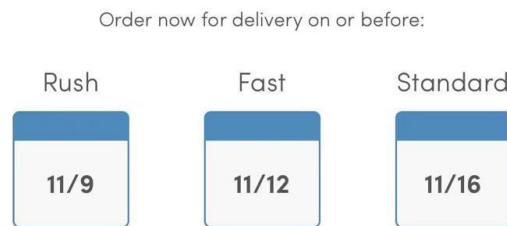
4.1. Research Background

We collaborated with a large national online retailer in the US, Collage.com, which specializes in customized photo products. Customers can design a collage by uploading photos and customizing the layout with the proprietary software tools on the website. Once a user creates the layout, they can purchase various types of customized printed products, such as blankets, photo-books, and

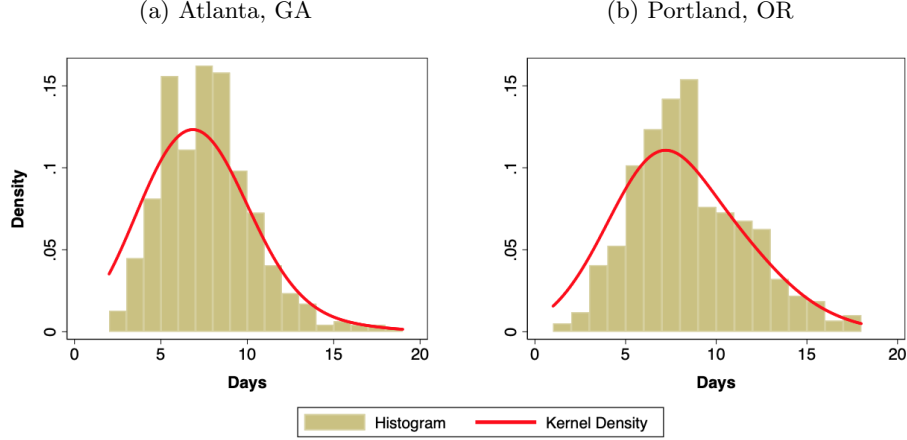
canvases. The layout is then printed on physical materials, and the physical products are shipped to the customers. Every year, tens of millions of customers visit the retailer’s website. Its annual revenue is more than 146 million dollars with a transaction volume that exceeds more than 4 million transactions per year. Given the volume of online traffic and sales, even a small improvement in the display of delivery speed information could lead to a large increase in the company’s current and future sales.

The online retailer operates in a **business-to-consumer** model and sells various types of customized prints directly to the customers. However, the retailer does not directly participate in the manufacturing process but contracts 13 offline vendors based in different locations in the US. Once an order is placed on the website, the retailer assigns the order to a designated vendor, who then manufactures the order based on the customer’s specifications. The retailer also partners with third-party carriers, including the U.S. Postal Service, UPS, DHL, and FedEx, to deliver products directly to customers after they are produced at a specific vendor. The retailer discloses the shipping speed estimates for any products of interest via the interface in Figure 2. Because the production and shipping processes are managed in a centralized fashion, the retailer has full visibility of the actual production and shipping time for each order. Due to uncertainties in production and delivery, there are large variations in the actual delivery speed; see Figure 3 for an example of the heterogeneity in the actual delivery time even within the same city. Therefore, customers will experience delays or early arrivals with any given delivery promise; see Figure 4 for an example. Given the spread, there is room for online retailers to optimize the delivery speed promise.

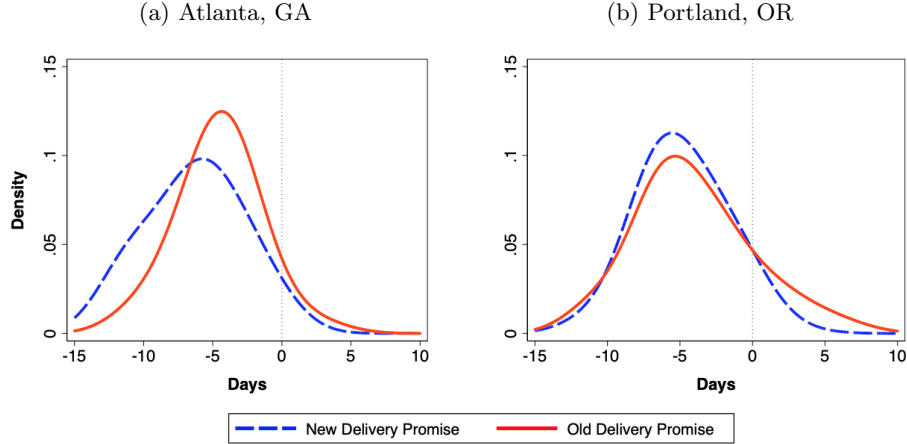
Figure 2 Disclosure of Delivery Speed Information



The retailer used to have a **heuristics** that disclosed the same delivery estimates for all cities in the US. We co-designed a new shipping disclosure policy with Collage that was implemented on December 9th, 2016. The new policy varied the delivery speed information across cities based on how close they were to the vendors. Those customers in cities that were closer to the vendors saw a faster delivery estimate on the website. The policy change was exogenous because it was applied at the national level and not tailored to any specific city. In addition, because the vendors are not specialized in printing customized photo products, and they cover a wide spectrum of businesses beyond Collage, their location choices are not strongly related with Collage’s specific

Figure 3 Examples of Actual Delivery Speed

These two graphs show the distributions of the actual delivery speed in days at the city level. We present Atlanta and Portland as examples. The actual delivery speed is calculated by the actual delivery date minus the order date. As shown, even within the same city, the actual delivery speed varies.

Figure 4 Examples of Delivery Delay

These two graphs show the distributions of delivery delays based on the new and the old delivery promises at the city level. We present Atlanta and Portland as examples. The delivery delay is calculated by the delivery date minus the promised date; hence, a negative number indicates the order was delivered earlier than promised, and a positive number indicates a delay. The delayed orders fall on the right side of the dotted line. As shown in the graphs, both the old estimate and the new estimate cannot accurately predict the actual delivery time, implying that considerable uncertainty exists in the delivery phase even within the same city. Moreover, it can be seen that the old estimate and the new estimate yield different delay distributions.

demand patterns. Therefore, this new policy change serves as a quasi-experiment and allows us to cleanly identify the effect of delivery speed information.

4.2. Data Description

We collected a granular transaction-level dataset for this study. First, for each order, we collected data on the purchasing value, the price of each purchased item, the profit margin, how much customers paid for shipping, and the product return decision (i.e., whether the product was returned). Second, for each order, we collected the delivery estimate using the website's event logs, and we

also collected the actual delivery speed using shipping carriers’ tracking data. Last, we used the post-order survey data to assess each customer’s satisfaction, including a binary “satisfied/not” response and a Net Promoter Score (NPS) of 1 to 10. Table 1 presents the summary statistics.

Our data spans from September 2016 to March 2017, including 212,340 transactions from 462 SKUs of photo products. We aggregated the data to create a panel structure over 7,090 cities across 24 weeks. We study three months before the policy change and three months after the policy change. Furthermore, we study five outcome variables that measure current sales and expenditures: the total number of orders per city, total profits per city, the purchasing value per order, the average item price per order, and the shipping expenditure per order.² We study three outcome variables that measure customers’ experience and indicate their future retention: product return rate, satisfaction rate, and NPS satisfaction score. In addition, we define the treatment variable as the number of days the retailer promised faster or slower shipping speed under the new disclosure policy relative to the delivery speed promise under the old disclosure policy.

Table 1 Summary Statistics

	Variable	Cities with Faster Delivery Promises (960 cities)		Control Cities (1,980 cities)		Cities of Slower Delivery Promises (4,150 cities)	
		Mean	Stdev	Mean	Stdev	Mean	Stdev
Before Policy Change	Orders	0.60	1.93	1.21	4.28	1.21	4.05
	Profits (\$)	49.41	160.59	99.02	363.37	95.13	316.60
	Order value (\$)	106.29	56.15	107.03	122.61	103.08	55.48
	Price (\$)	84.95	49.41	82.77	46.69	81.21	46.61
	Shipping expenditure (\$)	13.16	5.69	13.13	7.37	12.96	5.98
	Satisfied	0.93	0.25	0.91	0.26	0.89	0.30
	NPS	9.12	1.96	9.05	1.94	8.83	2.20
	Product return (%)	0.35	5.19	0.52	5.91	0.52	5.74
After Policy Change	Refund rate (%)	2.56	12.52	2.56	11.95	2.59	12.03
	Orders	0.55	1.71	1.11	4.00	1.05	3.60
	Profits (\$)	52.77	168.07	99.98	372.78	90.71	433.42
	Order value (\$)	125.87	97.66	116.65	80.17	108.68	63.45
	Price (\$)	100.68	54.74	91.41	49.73	85.68	49.48
	Shipping expenditure (\$)	15.56	8.88	14.60	7.44	13.93	7.38
	Satisfied	0.92	0.26	0.93	0.25	0.92	0.26
	NPS	9.16	1.87	9.13	1.84	9.06	1.92
	Product return (%)	0.60	6.95	0.54	5.89	0.60	6.32
	Refund rate (%)	2.30	12.04	2.21	11.19	2.25	11.36
	Delivery promise change (days)	1.46	0.81	-0.07	0.32	-1.70	0.94

5. Identification Strategy

In this section, we identify the causal effect of faster or slower delivery speed information on consumer behavior, including both purchasing and post-purchasing decisions. Specifically, in Section 5.1, we leverage the exogenous policy change by using a difference-in-differences regression to

² The shipping fee policy of Collage.com is illustrated in Figure 7. First, shipping is not free; even the standard ground shipping requires a charge. Second, shipping is more costly for more expensive products.

study purchasing decisions. In Section 5.2, we estimate the effect of delivery speed information on product returns and customer satisfaction using a transaction-level regression.

5.1. Difference-in-Differences Identification

To identify the effect of delivery on sales and expenditures, we estimate the following model specification with the city-level panel data,

$$Outcome_{it} = \alpha_i + \tau_t + \beta_1 Fast_i \times After_t + \beta_2 Slow_i \times After_t + e_{it}, \quad (1)$$

where i denotes the city, and t denotes the week. $Fast_i$ is the continuous treatment of how many days faster the retailer promises delivery in city i . $Slow_i$ is the continuous treatment of how many days slower the retailer promises. Variables α_i and τ_t are two fixed effects, where the former captures the time-invariant characteristics of city i , and the latter captures the time-variant factors that are common to all cities. For the time fixed effect τ_t , we include a set of weekly time dummies that can flexibly control for nonlinear trends. $After_t$ is a time dummy that indicates whether time t is after the policy change.³ $Outcome_{it}$ is the market outcome for city i in time t , which takes values in logged sales, profits,⁴ logged purchase value, logged price of purchased products, and logged expenditure in shipping. We validate the parallel trends assumption in Section 7.2.

5.2. Transaction-level Analysis

To identify the effect of delivery on product returns and customer satisfaction, we estimate the following model specification at the individual transaction level,

$$Outcome_{jt} = c + Fast_j + Slow_j + City_i + \tau_t + e_{it}, \quad (2)$$

where j denotes the individual transaction, i stands for the city, and t represents the week since the policy change. $Outcome_{jt}$ is the individual-level outcome for the individual transaction j in time t . $Outcome_{jt}$ takes values in the product return decision, customer satisfaction (binary score), and NPS score (0 to 10). Again, $Fast_j$ is the continuous treatment of how many days faster the retailer promises delivery for transaction j . $Slow_j$ is the continuous treatment of how many days slower the retailer promises. The transactions that have been promised a faster delivery time are more likely to experience a longer than expected waiting. Finally, we add two sets of dummy variables, $City_i$ and τ_t , to incorporate the city-level fixed effect and the week fixed effect.

In our dataset, returns and refunds are rare events; only a small portion of customers disclosed their satisfaction. This low-frequency problem is further exacerbated when we aggregate the

³ There is no need to include $after_t$ in the regression because the new policy was implemented at the beginning of a week. All its effects are incorporated by τ_t .

⁴ Because the profits are negative for some observations, we cannot use the logged value here.

individual-level data into the city-level data, where variations at the individual level are smoothed out. Thus, the analysis at the aggregate level easily fails to capture the subtle and useful information hidden in the data. To solve this issue, we employ Equation (2) to estimate the treatment effect on product returns and satisfaction at the individual transaction level.⁵

6. Estimation Results

We analyze customers' responses when the promised delivery estimates changed. Table 2 reports the average treatment effect of delivery speed estimates on sales, profits, order values, price of purchased items, and expenditures on shipping using Equation (1). Table 3 reports the average treatment effect of delivery speed estimates on product return and customer satisfaction using Equation (2).

We find a positive treatment effect for the faster delivery promise and a negative treatment effect for the slower delivery promise on sales, which suggests that delivery information directly influences customers' purchase decisions. Column (1) of Table 2 shows that sales increase by 0.73%⁶ with a one-day faster promise but decrease by 0.51% with a one-day slower promise. This result confirms our hypothesis that a faster delivery speed promised to customers will lead to more purchases. Additionally, a one-day faster promise increases profits by \$1.77, or 2.0% per order, while a one-day slower promise decreases profits by \$2.46, or 2.7% per order.⁷ Further, an aggressive disclosure contributes to revenue not only through higher order volumes but also through greater order values. The order value increases at a rate of 3.5% per day with a faster delivery promise; on the other hand, it decreases at a rate of 3.1% per day with a slower delivery promise. Moreover, we identify that most changes in order values come from changes in the purchased item's price. A one-day faster delivery promise increases the average price of purchased items by 3.2% and increases consumers shipping expenditures by 2.0%; in contrast, a one-day slower promise decreases these two categories by 4.3% and 3.1%, respectively. Therefore, Hypothesis 1 is supported.

However, there are consequences to faster delivery promises. We find a positive and significant treatment effect of a faster delivery promise on product return. As shown in column (1) of Table 3, a one-day faster promise increases the return rate by 0.04% in level according to the linear regression model. This amount translates into a 6.4% increase in percentage given that the overall return rate in this subsample is 0.62%.⁸ We also applied the probit and logit models, which yield very

⁵ We would like to point out that this change comes at a cost. When using Equation (2), the DID framework is lost because we cannot compare customer decisions before and after the policy change. Nonetheless, Equation (2) can better capture the variations of the outcome variables in response to the delivery promise changes.

⁶ Because the outcome variables are logged, the percentage change in sales equals $e^{0.0073} - 1 = 0.73\%$. Similar transformations are used for order values, prices of purchased items and shipping expenditure.

⁷ The percentages are calculated by $\frac{\Delta \text{profit}}{\text{average profit}}$, where the denominator is the average profit before the policy change.

⁸ The percentage is calculated by $\frac{\Delta \text{return rate}}{\text{average return rate}}$, according to which we get $\frac{0.04}{0.62} = 6.4\%$.

Table 2 Impact of Delivery Speed Promise on Sales and Spending

	Log orders (1)	Profits (2)	Log order value (3)	Log item price (4)	Log shipping expense (5)
Fast $\times$ After	0.0073** (0.0030)	176.7* (94.7)	0.0347*** (0.0125)	0.0316** (0.0148)	0.0200* (0.0108)
Slow $\times$ After	-0.0051*** (0.0018)	-245.9*** (56.5)	-0.0310*** (0.0053)	-0.0436*** (0.0067)	-0.0313*** (0.0044)
City FE	Yes	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	170,160	170,160	60,748	60,748	60,748
R^2	0.243	0.093	0.021	0.015	0.047

This table reports the estimated coefficients and cluster-robust standard errors (in parentheses) in Equation (1). The coefficients for the number of orders, profits, the purchasing value per order, the average price per item, and consumers' expenditures in shipping are presented in columns (1)–(5), respectively. The standard errors are clustered at the city level. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

similar estimates. These results verify the concern that without changing the physical delivery, an aggressive promise causes more delivery delays and thus leads to higher returns. On the other hand, there is no statistically significant evidence showing that a slower delivery promise will decrease returns. Therefore, Hypothesis 2 is partially supported.

We also test the impact on customer satisfaction, which is shown in columns (4) - (7) of Table 3. Here, we have two different indicators that reflect customer satisfaction: *Satisfied* and *NPS*. Both indicators' results convey the same message: a slower delivery promise lowers customer satisfaction, and a faster promise has no statistically significant effect. Specifically, according to column (4) of Table 3, a one-day slower promise decreases the satisfaction rate by 0.56%. Note that these results should be interpreted with caution: only 8.5% of total customers disclosed their satisfactions. This low response rate could have caused a sample selection problem, resulting in potentially biased estimates. A slower delivery promise may have disappointed the small group of customers who chose to express their unhappiness, but its effect on the whole population remains to be studied.

Table 3 Impact of Delivery Speed Promise on Return and Satisfaction

	Return			Satisfied			NPS
	(1) linear	(2) probit	(3) logit	(4) linear	(5) probit	(6) logit	(7) linear
Fast	0.0004* (0.0002)	0.0183** (0.0083)	0.0511** (0.0228)	-0.0020 (0.0023)	-0.0058 (0.0134)	-0.0111 (0.0270)	0.0044 (0.0170)
Slow	-0.0000 (0.0002)	-0.0030 (0.0097)	-0.0084 (0.0273)	-0.0070* (0.0037)	-0.0585*** (0.0130)	-0.1150*** (0.0249)	-0.0682*** (0.0251)
City FE	Yes	No	No	Yes	No	No	Yes
Week FE	Yes	No	No	Yes	No	No	Yes
Observations	84,449	87,059	87,059	5,390	7,436	7,436	5,390
R^2	0.0088	0.0008	0.0008	0.0077	0.0050	0.0050	0.0271

This table reports the estimated coefficients and standard errors (in parentheses) in Equation (2). Regressions are run at the individual order level. The estimates for whether the order is returned using linear, probit, and logit models are presented in columns (1)–(3). The estimates for whether the customer is satisfied using linear, probit, and logit models are presented in columns (4)–(6). The estimates for the satisfaction score are presented in column (7). For linear models, city- and week-fixed effects are included, and standard errors are clustered at the city level. For nonlinear models, we did not run the regression with fixed effects, as it suffers from an incidental parameter problem (Heckman 1987, Cameron and Trivedi 2005). Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

7. Robustness Check

In this section, we report several robustness tests to validate our results. Specifically, we show that the change in the delivery promise is independent of the physical delivery speed, and we test the parallel trends assumption and alternative empirical specifications.

7.1. Change in Actual Delivery Speed

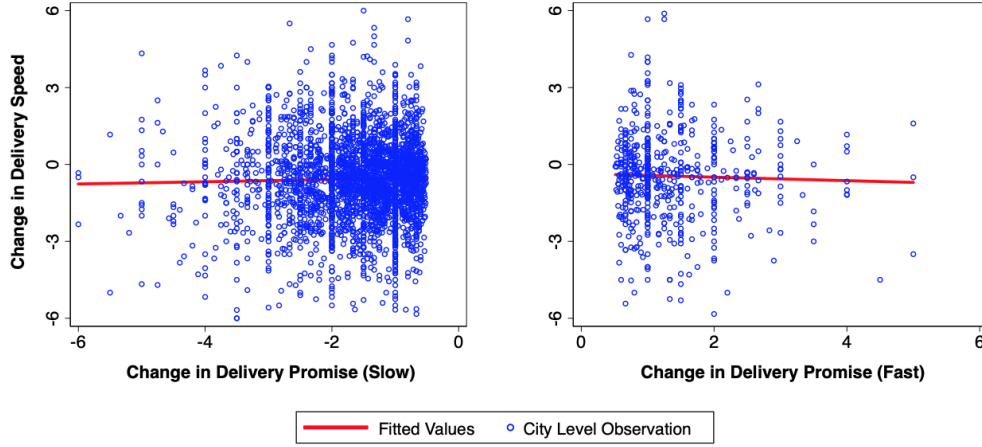
Our results rely on the assumption that the changes in the delivery promise do not depend on the changes in the delivery speed. If this assumption is violated, we would risk misattributing the effect of the delivery speed to the effect of the delivery promise. However, this assumption is likely to hold for two reasons. First, the delivery speed is mainly determined by the physical delivery setups, such as the transportation mode and fleet, which may take years to build. Our analysis focuses on a six-month time window around the policy change. Within such a time period, it is unlikely for the physical delivery to be significantly different,⁹ therefore, all changes in the delivery promise should be primarily informational. Second, even if there are variations in the physical delivery setups, as long as the change in delivery promise is independent of the change in actual delivery, we can still obtain an unbiased causal estimate. In other words, we need to rule out the possibility that the change in delivery promise is confounded with a change in physical delivery. This is unlikely to be true because our experiment varied the speed information based primarily on the cities' pre-determined geographical distance to the existing vendors. The static nature of geographical distances precludes our treatment from being connected to dynamic and forward-looking factors, such as future changes in physical setups.

In addition, we directly test this assumption by regressing the change in delivery speed on the change in delivery promise for the treatment group with an aggressive delivery promise and the treatment group with a conservative delivery promise, separately,

$$\Delta\text{Delivery Speed}_i = c + \beta\Delta\text{Delivery Promise}_i, \quad (3)$$

where $\Delta\text{Delivery Speed}_i$ is the change in delivery speed for city i before and after the policy, and similarly, $\Delta\text{Delivery Promise}_i$ is the change in delivery promise. Figure 5 visually plots the regressions, where the fitted regressions are almost flat, and the data points are evenly distributed along both sides of the regression lines. Table 4 shows that both β_{Slow} and β_{Fast} are not significant. These empirical results confirm that the irrelevance assumption is supported by the data.

⁹ This does not mean that there is no change in the observed delivery speed, as the actual delivery processes are affected by other random factors such as the weather condition.

Figure 5 Actual Delivery Speed Unaffected by the Delivery Speed Promise

These two graphs plot the change in delivery promise relative to the change in delivery speed for the treatment group with faster delivery promises and the treatment group with slower delivery promises, respectively. The solid red lines represent the fitted regressions. The scatter plots represent the city-level observations. The change in delivery promise appears to be random with respect to the change in delivery speed. The low R^2 indicates that only 0.01% of the observed variations in $\Delta\text{Delivery Speed}_i$ can be attributed to variations in $\Delta\text{Delivery Promise}_i$.

Table 4 Delivery Speed Promise Not Affect the Actual Delivery Speed

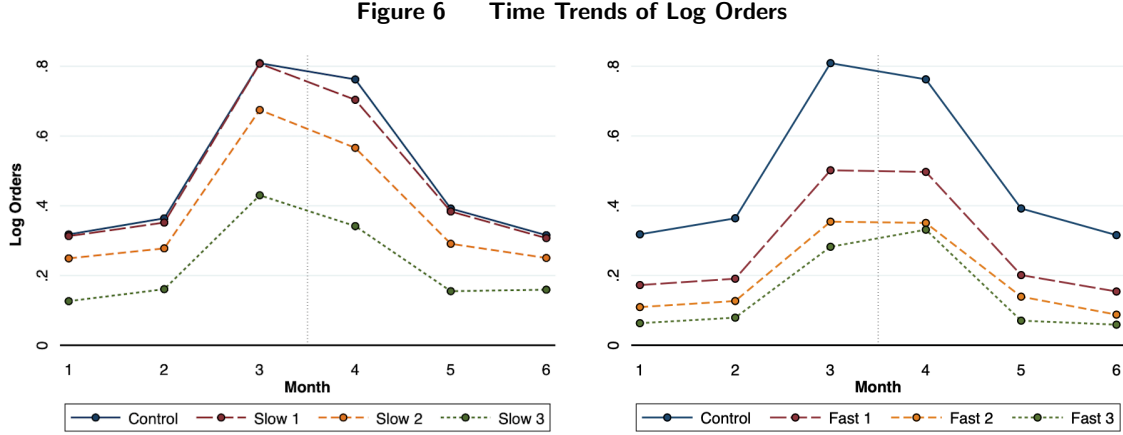
	Coefficient Value	Standard Error	p -value	R^2
β_{Slow}	0.0224	0.0384	0.560	0.0001
β_{Fast}	-0.0265	0.1023	0.795	0.0001

7.2. Parallel Trends

The key identification assumption for the DID strategy is the **parallel trends assumption**: before the policy change, the treatment and control groups would follow the same time trend. We perform empirical tests to confirm that there are no pre-treatment differences in sales trends between the treatment and control cities. Because our treatments are continuous, we discretized the treatments and split them into a finite number of groups, as shown in Table 5. Figure 6 presents the time trends of the log orders. The left graph shows that the slow groups evolved in parallel with the control group before the policy change. After the policy change, however, the slow groups' sales dropped much more than the control group, indicating that a slower delivery promise has a negative impact on sales. We can observe a similar pattern for the faster delivery promise group: the trend evolved in parallel before the policy change, and the fast group's sales dropped less than the control group.

Table 5 Discretizing the Treatments

Group	Slow 3	Slow 2	Slow 1	Control	Fast 1	Fast 2	Fast 3
$\Delta\text{Delivery Promise}$	$(-6, -2.5]$	$(-2.5, -1.5]$	$(-1.5, -0.5]$	$(-0.5, 0.5)$	$[0.5, 1.5)$	$[1.5, 2.5)$	$[2.5, 6)$



These two graphs plot the time trends of the discretized treatment groups and the control group. Although our data has a weekly frequency, to smooth out fluctuations, we average and plot log orders for each month.

7.3. Alternative Empirical Specification

We demonstrate that our main results in Table 2 are robust to model specifications. The DID identification strategy has taken several flexible forms in the literature. We use an alternative DID model, as specified in Fisher et al. (2016) and Cui et al. (2019a),

$$Outcome_{it} = c + Fast_i + Slow_i + Week_t + \beta_1 Fast_i \times After_t + \beta_2 Slow_i \times After_t + e_{it}. \quad (4)$$

The estimation results are reported in Table 6. Comparing the estimates between Tables 2 and 6, we can see that the two specifications yield highly robust results.

Table 6 Robust Specifications

	Log orders (1)	Profits (2)	Log order value (3)	Log item price (4)	Log shipping expense (5)
Fast	-0.1360*** (0.0061)	-3994.2*** (246.3)	0.0002 (0.0083)	0.0250** (0.0101)	0.0023 (0.0067)
Slow	-0.0488*** (0.0040)	-1787.7*** (172.3)	-0.0131*** (0.0037)	-0.0054 (0.0047)	-0.0076** (0.0029)
Fast $\times$ After	0.0073** (0.0030)	176.7* (94.7)	0.0426*** (0.0115)	0.0376*** (0.0137)	0.0270*** (0.0099)
Slow $\times$ After	-0.0051*** (0.0018)	-245.9*** (56.5)	-0.0367*** (0.0051)	-0.0508*** (0.0065)	-0.0357*** (0.0042)
Week FE	Yes	Yes	Yes	Yes	Yes
Observations	170,160	170,160	60,748	60,748	60,748
R^2	0.151	0.063	0.046	0.017	0.046

This table reports the estimated coefficients and cluster-robust standard errors (in parentheses) in Equation (4). The coefficients for the number of orders, profits, the purchasing value per order, the average price per item, and consumers' expenditures in shipping are listed in columns (1)–(5), respectively. The standard errors are clustered at the city level. Significance at * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

8. Conclusions

By designing and studying a new shipping disclosure policy for Collage.com, this paper takes one of the first steps toward quantifying the economic impact of disclosing aggressive or conservative

shipping estimates to customers. We demonstrate that the online retailer can significantly benefit from managing the disclosure of delivery speed information. It is an effective way to boost sales, profits, and customer satisfaction as well as to reduce product returns. Moreover, we highlight the trade-offs of this information: executing an aggressive disclosure can increase sales at the cost of more product returns and perhaps a lower retention rate in the future.

These insights, derived in our context, can be generalized across many situations. Given the variations in the physical delivery speed, there is always value to customize the speed promised to customers. Companies competing in the online retailing space can leverage this study by tailoring their information disclosure policies as a function of individual characteristics. For example, retailers can use data-driven policies, such as machine learning, to prescribe the delivery estimates in light of the ever-changing context of each product, each geographical region, and each individual customer. With the rapid advancement in firms capabilities to process a large amount of real-time consumer behavior data, we envision the ability of firms to optimize and share the delivery speed information in an instantaneous and personalized fashion.

The retailer that we partnered with has found value in and is ready to implement the recommendations of our work. In the words of the retailer’s CEO, this project “provides us insights that help us discuss and subsequently implement changes in our shipping policy, as well as test alternative shipping information models in our company.” In the face of the emphasis on delivery speed and stability, the significance of a firm’s information sharing strategies has been intensified with their ever-increasing impact on consumer decisions. We hope that this paper will serve as a stepping stone for future research to explore how to manage delivery speed information in online retail.

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Appendices

Figure 7 Collage's Shipping Policy

Product		Standard Shipping	Fast Shipping	Rush Shipping	International Shipping	Notes
Fleece Blankets		\$15.98	\$27.94	\$33.94	\$39.99	Extra shipping charge for each additional item: \$9.99 Expedited and international shipping not available for king size blankets
Sherpa Blankets		\$15.98	\$27.94	\$33.94	\$39.99	Extra shipping charge for each additional item: \$9.99
Woven Image Blankets		\$15.98	N/A	N/A	U.S. Shipping Only	Extra shipping charge for each additional item: \$9.99
Photo Books	Softcover	\$5.99	\$17.95	\$23.95	\$8.99	Extra shipping charge for each additional item: \$5.99
	Small Hardcover (8"x8")	\$5.99	\$17.95	\$23.95	\$8.99	Extra shipping charge for each additional item: \$5.99
	Hardcover (11½"x7½" & 11½"x8½")	\$9.99	\$21.95	\$27.95	\$12.99	Extra shipping charge for each additional item: \$9.99
	Large Hardcover (12"x12" & 14"x10½")	\$15.98	\$27.94	\$33.94	\$18.98	Extra shipping charge for each additional item: \$9.99
Canvas Prints / Gallery Wrapped Canvases	Small 5x7 & 6x6	\$9.98	\$21.94	\$27.94	U.S. Shipping Only	Does not ship to P.O. boxes. Extra shipping charge for each additional item: \$3.99
	Medium 8x10, 10x10, 12x12, 11x14, 10x20, 12x18, 16x16, 16x20, 16x24, 20x20, 20x24 & 24x24	\$15.98	\$27.94	\$33.94	U.S. Shipping Only	Does not ship to P.O. boxes. Extra shipping charge for each additional item: \$9.99
	Large 20x30, 24x30 & 24x36	\$19.98	\$31.94	\$37.94	U.S. Shipping Only	Does not ship to P.O. boxes. Extra shipping charge for each additional item: \$13.99
	Extra Large 12x36, 30x40, 32x48, 30x30, 36x36 & 20x60	\$25.98	\$37.94	\$43.94	U.S. Shipping Only	Does not ship to P.O. boxes. Extra shipping charge for each additional item: \$19.99
Framed Canvas Prints	Medium 12x12, 11x14, 12x18, 16x20	\$15.98	\$27.94	\$33.94	U.S. Shipping Only	Does not ship to P.O. boxes. Extra shipping charge for each additional item: \$9.99
	Large 20x30	\$15.98	\$27.94	\$33.94	U.S. Shipping Only	Does not ship to P.O. boxes. Extra shipping charge for each additional item: \$9.99
	Extra Large 12x36	\$15.98	\$27.94	\$33.94	U.S. Shipping Only	Does not ship to P.O. boxes. Extra shipping charge for each additional item: \$9.99
Throw Pillows	With Insert	\$9.98	\$21.94	\$27.94	U.S. Shipping Only	Extra shipping charge for each additional item: \$5.99
	Cover Only	\$9.98	\$21.94	\$27.94	U.S. Shipping Only	No extra per-item charge for orders of up to 9 items. Quantities of 10 or more may result in additional shipping charges.

The detailed shipping policy can be found at <https://www.collage.com/shippolicy>.